## INTRODUCTION

The branch of electronics, which deals with digital circuits, is called digital electronics. Over the past several decades, digital electronics have been utilized in the design and manufacturing of various industrial, commercial and household electronic gadgets. Due to the proliferation of digital electronics, it is very important to inculcate the basic knowledge of digital electronics to develop conceptual knowledge and practical experience among the stakeholders.

Electronic systems can be classified into two types of systems in which the mode of electron transfer from one end to another end differs. They are,

1. Analog system
2. Digital system

This Chapter covers the basic concepts in digital electronics.

**Who is the inventor of logic gates?**

Walther Bothe, inventor of the coincidence circuit, got part of the 1954 Nobel Prize in Physics, for the first modern electronic AND gate in 1924. Konrad Zuse designed and built electromechanical logic gates for his computer Z1 (1935-38)

## 8.1. ANALOG AND DIGITAL SIGNALS

### i) Analog Signals

A continuously varying signal(voltage or current) is called as an analog signal.

**Example:** Sinusoidal waves.

A sample of analog signal that varies with time is shown in Figure 8.1.

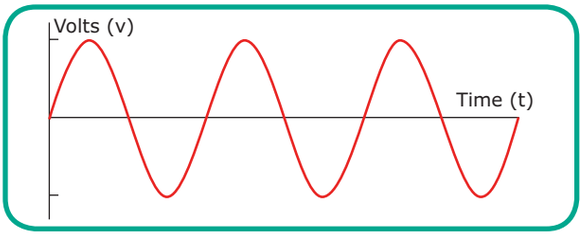



Figure8.1 Representation of an Analog signal

### ii) Digital Signal

A signal (voltage or current) that can have only two discrete values is called a digital signal. Example: Square wave. The digital waveform is shown in Figure 8.2.

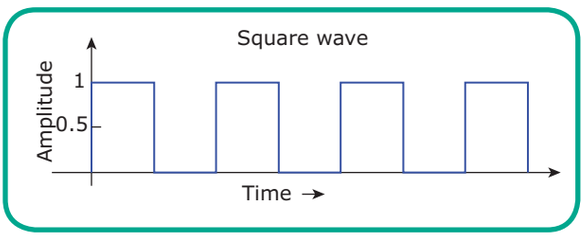



Figure 8.2 Representation of Digital Signal

Digital operations have two states (i.e. ON or OFF) and hence it is more simple and reliable than many valued analog operations.

## 8.2 DIGITAL CIRCUITS

An electronic circuit that handles only a digital signal is called a digital circuit.

**Example:** Digital calculator, Digital computer

The digital operation is a two state operation (i.e. ON or OFF, 1 or 0) and therefore a digital circuit uses only two digits 1 and 0 in the binary number system. In order to understand the concepts in digital circuits, first we discuss



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